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Method for manufacturing glass panel unit

Abstract

A method for manufacturing a glass panel unit includes a glue arrangement step, a pillar placement step, an assembly forming step, a bonding step, an evacuation step, and a sealing step. The evacuation step reduces pressure in an internal space by exhausting, with predetermined suction power, a gas from the internal space via a valve and an exhaust port. The valve includes a first valve allowing the gas to flow through a first channel area and a second valve allowing the gas to flow through a second channel area larger than the first channel area. The evacuation step includes a first evacuation step to be performed first and a second evacuation step to be performed next. In the first evacuation step, the first valve is opened and the second valve is closed. In the second evacuation step, the first valve is closed and the second valve is opened.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6984159	12/2005	Kado et al.	N/A	N/A
2004/0157010	12/2003	Minaai et al.	N/A	N/A
2015/0068666	12/2014	Abe et al.	N/A	N/A
2018/0283087	12/2017	Abe et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
H11-311069	12/1998	JP	N/A
2001-064043	12/2000	JP	N/A
2001-351525	12/2000	JP	N/A
2016-029020	12/2015	JP	N/A
2018-012637	12/2017	JP	N/A
10-2009-0069736	12/2008	KR	N/A

OTHER PUBLICATIONS

CN 104961358, machine translation (Year: 2015). cited by examiner
Communication pursuant to Rule 114(2) EPC dated Oct. 29, 2021, issued in the corresponding European Patent Application No. 19827448.2. cited by applicant
International Search Report and Written Opinion issued in International Patent Application No. PCT/JP2019/020555, dated Jul. 16, 2019; with partial English translation. cited by applicant

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Background/Summary

CROSS-REFERENCE OF RELATED APPLICATIONS
(1) This application is the U.S. National Phase under 35 U.S.C. § 371 of International Patent Application No. PCT/JP2019/020555, filed on May 23, 2019, which in turn claims the benefit of Japanese Application No. 2018-121338, filed on Jun. 26, 2018, the entire disclosures of which Applications are incorporated by reference herein.
TECHNICAL FIELD
(2) The present invention generally relates to a method for manufacturing a glass panel unit, and more particularly relates to a method for manufacturing a glass panel unit including exhausting a gas from an internal space of an assembly including a first panel, a second panel, a hot glue, and pillars.
BACKGROUND ART
(3) Patent Literature 1 discloses a multi-pane glazing unit. The multi-pane glazing unit of Patent Literature 1 includes: a first panel; a second panel arranged to face the first panel; and a sealant that hermetically bonds the first and second panels together. The multi-pane glazing unit further includes a plurality of spacers arranged in an internal space, which is sealed by the first panel, the second panel, and the sealant to be an evacuated space, to conic into contact with the first and second panels.
(4) To manufacture the known multi-pane glazing unit, the sealant and the spacers are arranged on either the first panel or the second panel so that the first and second panels are hermetically bonded together with the sealant, This increases the chances of the spacers arranged on the first panel or the second panel being displaced when a gas is exhausted from the internal space.
CITATION LIST
Patent Literature
(5) Patent Literature 1: JP H11-311069 A
SUMMARY OF INVENTION
(6) It is therefore an object of the present invention to provide a method for manufacturing a glass panel unit which reduces the chances of the pillars arranged on the first panel or the second parcel being displaced when the gas is exhausted from the internal space.
(7) A method for manufacturing a glass panel unit according to an embodiment of the present disclosure includes a glue arrangement step, a pillar placement step, an assembly forming step, a bonding step, an evacuation step, and a sealing step.
(8) The glue arrangement step includes arranging a hot glue on either a first panel including a first glass pane or a second panel including a second glass pane.
(9) The pillar placement step includes placing pillars on either the first panel or the second panel.
(10) The assembly forming step includes forming an assembly including the first panel, the second panel, the hot glue, and the pillars and having an exhaust port provided through at least one of the first panel, the second panel, or the hot glue by arranging the second panel such that the second panel faces the first panel.
(11) The bonding step includes heating the assembly to melt the hot glue, bonding the first panel and the second panel together with the hot glue thus melted, and thereby forming an internal space surrounded, except the exhaust port, with the first panel, the second panel, and the hot glue melted.
(12) The evacuation step includes reducing pressure in the internal space by evacuation that

involves exhausting, with predetermined suction power, a gas from the internal space via parallel channels and the exhaust port.
(13) The sealing step includes creating a hermetically sealed evacuated space by closing the exhaust port and sealing the internal space while maintaining a reduced pressure in the internal space.
(14) The parallel channels include: a plurality of channels that are connected together in parallel; and one or two or more on-off valves provided for one or two or more channels out of the plurality of channels. A total channel area of the parallel channels is changeable according to a combination of opened and closed states of the on-off valves.
(15) The evacuation step includes: a first evacuation step to be performed first; and a second evacuation step to be performed next to the first evacuation step.
(16) The first evacuation step includes allowing the gas to flow while setting the total channel area of the parallel channels at a first total channel area.
(17) The second evacuation step includes allowing the gas to flow while setting the total channel area of the parallel channels at a second total channel area that is larger than the first total channel area.
(18) A method for manufacturing a glass panel unit according to another embodiment of the present disclosure includes a glue arrangement step, a pillar placement step, an assembly forming step, a bonding step, an evacuation step, and a sealing step.
(19) The glue arrangement step includes arranging a hot glue on either a first panel including a first glass pane or a second panel including a second glass pane.
(20) The pillar placement step includes placing pillars on either the first panel or the second panel.
(21) The assembly forming step includes forming an assembly including the first panel, the second panel, the hot glue, and the pillars and having an exhaust port provided through at least one of the first panel, the second panel, or the hot glue by arranging the second panel such that the second panel faces the first panel.
(22) The bonding step includes heating the assembly to melt the hot glue, bonding the first panel and the second panel together with the hot glue thus melted, and thereby forming an internal space surrounded, except the exhaust port, with the first panel, the second panel, and the hot glue melted.
(23) The evacuation step includes reducing pressure in the internal space by evacuation that involves exhausting a gas, using a pump configured to suction the gas with predetermined suction power, from the internal space via a predetermined channel and the exhaust port.
(24) The sealing step includes creating a hermetically sealed evacuated space by closing the exhaust port and sealing the internal space while maintaining a reduced pressure in the internal space.
(25) The predetermined channel includes: an exhaust channel located on a path connecting the pump to the exhaust port and having an exhaust on-off valve; and an air-introducing channel located on a path connecting the pump to the air and having an air-introducing on-off valve.
(26) The evacuation step includes: a first evacuation step to be performed first; and a second evacuation step to be performed next to the first evacuation step.
(27) The first evacuation step includes opening the exhaust on-off valve and opening the air-introducing on-off valve.
(28) The second evacuation step includes opening the exhaust on-off valve and closing the air-introducing on-off valve.
(29) A method for manufacturing a glass panel unit according to still another aspect of the present disclosure includes a glue arrangement step, a pillar placement step, an assembly forming step, a bonding step, an evacuation step, and a sealing step.
(30) The glue arrangement step includes arranging a hot glue on either a first panel including a first glass pane or a second panel including a second glass pane.
(31) The pillar placement step includes placing pillars on either the first panel or the second panel.

(32) The assembly forming step includes forming an assembly including the first panel, the second panel, the hot glue, and the pillars and having an exhaust port provided through at least one of the first panel, the second panel, or the hot glue by arranging the second panel such that the second panel faces the first panel.
(33) The bonding step includes heating the assembly to melt the hot glue, bonding the first panel and the second panel together with the hot glue thus melted, and thereby forming an internal space surrounded, except the exhaust port, with the first panel, the second panel, and the hot glue melted.
(34) The evacuation step includes reducing pressure in the internal space by evacuation that involves exhausting, with predetermined suction power, a gas from the internal space via a valve and the exhaust port.
(35) The sealing step includes creating a hermetically sealed evacuated space by closing the exhaust port and sealing the internal space while maintaining a reduced pressure in the internal space.
(36) The evacuation step includes: a first evacuation step to be performed first; and a second evacuation step to be performed next to the first evacuation step. The first evacuation step and the second evacuation step each include controlling opened and closed states of the valve.
(37) A first ratio of a valve open duration during the first evacuation step to an overall running time of the first evacuation step is smaller than a second ratio of a valve open duration during the second evacuation step to an overall running time of the second evacuation step. The valve open duration is a period of time for which the valve is opened.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a schematic cross-sectional view of a glass panel unit according to a first embodiment;
(2) FIG. **2** is a partially cutaway, schematic plan view of the glass panel unit;
(3) FIG. **3** is a schematic cross-sectional view of an assembly of the glass panel unit;
(4) FIG. **4** is a partially cutaway, schematic plan view of the assembly;
(5) FIG. **5** illustrates a method for manufacturing the glass panel unit;
(6) FIG. **6** illustrates the method for manufacturing the glass panel unit;
(7) FIG. **7** illustrates the method for manufacturing the glass panel unit;
(8) FIG. **8** illustrates the method for manufacturing the glass panel unit;
(9) FIG. **9**A illustrates an exemplary exhaust system for use in the method for manufacturing the glass panel unit;
(10) FIG. **9**B illustrates another exemplary exhaust system for use in the method for manufacturing the glass panel unit;
(11) FIG. **10** is a timing diagram showing how pressure in an internal space changes with time in the method for manufacturing the glass panel unit;
(12) FIG. **11** is a timing diagram showing how pressure in an internal space changes with time in a comparative example;
(13) FIG. **12** illustrates an exemplary exhaust system for use in a method for manufacturing a glass panel unit according to a second embodiment;
(14) FIG. **13** illustrates an exemplary exhaust system for use in a method for manufacturing a glass panel unit according to a third embodiment;
(15) FIG. **14** is a timing diagram showing how pressure in an internal space changes with time in the method for manufacturing the glass panel unit;
(16) FIG. **15**A is a partially cutaway, schematic plan view of an assembly according to a fourth embodiment, and
(17) FIG. **15**B is a partially cutaway, schematic plan view of an assembly according to another

example of the fourth embodiment.

DESCRIPTION OF EMBODIMENTS

(18) The first to fourth embodiments to be described below generally relate to a method for manufacturing a glass panel unit, and more particularly relate to a method for manufacturing a glass panel unit including exhausting a gas from an internal space of an assembly including a first panel, a second panel, a hot glue, and pillars.

(19) A method for manufacturing a glass panel unit according to a first embodiment will be described with reference to FIGS. **1-11**.

(20) FIGS. **1** and **2** illustrate a glass panel unit (which is a glass panel unit as a final product) **10** according to the first embodiment. The glass panel unit **10** according to the first embodiment is a thermal insulation glazing unit. The thermal insulation glazing unit is a type of a multi-pane glazing unit (multi-pane glass panel unit) including at least a pair of glass panels and having an evacuated space (which may be a vacuum space) between the pair of glass panels.

(21) The glass panel unit **10** according to the first embodiment includes a first panel **20**, a second panel **30**, a seal **40**, an evacuated space **50**, a gas adsorbent **60**, a plurality of pillars **70**, and a closing member **80**.

(22) The glass panel unit **10** is obtained by processing an assembly **100** as an intermediate product shown in FIGS. **3** and **4**.

(23) The assembly **100** includes at least the first panel **20**, the second panel **30**, a hot glue (including a frame member **410** and a partition **420** to be described later), and the pillars **70**, and has an exhaust port **700**. Specifically, according to the first embodiment, the assembly **100** includes the first panel **20**, the second panel **30**, the frame member **410** and partition **420** serving as a hot glue, an internal space **500**, an air passage **600**, the exhaust port **700**, the gas adsorbent **60**, and the plurality of pillars **70**.

(24) The first panel **20** includes a first glass pane **21** defining the planar shape of the first panel **20** and a coating **22**.

(25) The first glass pane **21** is a rectangular flat plate and has a first surface (the lower surface shown in FIG. **3**) and a second surface (the upper surface shown in FIG. **3**) on both sides in the thickness direction. The first and second surfaces are parallel to each other. The first glass pane **21** with the rectangular shape may have a length of about 1360 mm to about 2350 mm each side, and a thickness of about 1 mm to about 20 mm, for example. However, these numerical values are only examples and should not be construed as limiting. Examples of materials for the first glass pane **21** include soda lime glass, high strain point glass, chemically tempered glass, alkali-free glass, quartz glass, Neoceram, and thermally tempered glass.

(26) The coating **22** is formed on the first surface of the first glass pane **21**. In the first embodiment, a so-called “Low-E (low-emissivity)” film is formed as the coating **22** on the first surface of the first glass pane **21**. The coating **22** does not have to be a Low-E film but may also be any other film with desired physical properties. Optionally, the first panel **20** may consist of the first glass pane **21**. In other words, the first panel **20** includes at least the first glass pane **21**. In the first embodiment, the Low-E film is formed by sputtering as the coating **22** on the first surface of the first glass pane **21**.

(27) The second panel **30** includes a second glass pane **31** defining the planar shape of the second panel **30**. The second glass pane **31** is a rectangular flat plate and has a first surface (the upper surface shown in FIG. **3**) and a second surface (the lower surface shown in FIG. **3**) on both sides in the thickness direction. The first and second surfaces of the second glass pane **31** are parallel to each other and are flat surfaces.

(28) The second glass pane **31** may have the same planar shape and same planar dimensions as the first glass pane **21**. That is to say, the second panel **30** may have the same planar shape as the first panel **20**. Also, the second glass pane **31** may be as thick as the first glass pane **21**. Examples of materials for the second glass pane **31** include soda lime glass, high strain point glass, chemically

tempered glass, alkali-free glass, quartz glass, Neoceram, and thermally tempered glass.
(29) The second panel **30** may consist of the second glass pane **31**. That is to say, the second glass pane **31** may be the second panel **30** itself. Optionally, the second panel **30** may have a coating on either surface thereof. The coating is a film with desired physical properties (such as an infrared reflective film). In that case, the second panel **30** is made up of the second glass pane **31** and the coating. In short, the second panel **30** includes at least the second glass pane **31**.
(30) The second panel **30** is arranged to face the first panel **20**. Specifically, the first panel **20** and the second panel **30** are arranged such that the first surface of the first glass pane **21** and the first surface of the second glass pane **31** are parallel to each other and face each other.
(31) The frame member **410** is arranged between the first panel **20** and the second panel **30** to hermetically bond the first panel **20** and the second panel **30** together. Thus, an internal space **500** is formed which is surrounded with the frame member **410**, the first panel **20**, and the second panel **30**.
(32) The frame member **410** is made of a hot glue (which is a first hot glue with a first softening point). The first hot glue may be a glass frit, for example. The glass frit may be a low-melting glass frit, for example. Examples of the low-melting glass frits include a bismuth-based glass frit, a lead-based glass frit, and a vanadium-based glass frit.
(33) The frame member **410** has a rectangular frame shape. The planar shape of the frame member **410** may be substantially the same as that of the first glass pane **21** and the second glass pane **31**. However, the frame member **410** has smaller planar dimensions than the first glass pane **21** and the second glass pane **31**. The frame member **410** is formed along the outer periphery of the upper surface of the second panel **30** (i.e., the first surface of the second glass pane **31**). That is to say, the frame member **410** is formed to surround almost the entire region on the second panel **30** (i.e., the first surface of the second glass pane **31** almost entirely).
(34) The first panel **20** and the second panel **30** are hermetically bonded together with the frame member **410** by once melting the first hot glue of the frame member **410** at a predetermined temperature (first melting temperature) Tm**1** (see FIG. **8**) which is equal to or higher than the first softening point.
(35) The partition **420** is arranged in the internal space **500**. The partition **420** partitions the internal space **500** into a hermetically sealed space, i.e., a first space **510** which will be hermetically sealed to form an evacuated space **50** when the glass panel unit **10** is completed, and a gas exhausting space, i.e., a second space **520** communicating with the exhaust port **700**. The partition **420** is provided closer to a first end (i.e., the right end in FIG. **4**) along the length of the second panel **30** (i.e., the rightward/leftward direction in FIG. **4**) than to a middle of the length of the second panel **30** such that the first space **510** becomes larger than the second space **520**.
(36) The partition **420** is made of a hot glue (i.e., a second hot glue having a second softening point). The second hot glue may be a glass frit, for example. The glass frit may be a low-melting glass frit, for example. Examples of the low-melting glass frits include a bismuth-based glass frit, a lead-based glass frit, and a vanadium-based glass frit. The second hot glue is the same as the first hot glue. The second softening point is equal to the first softening point.
(37) An air passage **600** that allows the first space **510** to communicate with the second space **520** is provided through a part of the partition **420**.
(38) The exhaust port **700** is a hole that connects the second space **520** to an external environment. The exhaust port **700** is used to exhaust a gas from the first space **510** through the second space **520** and the air passage **600**. The exhaust port **700** is provided through the second panel **30** to connect the second space **520** to the external environment. Specifically, the exhaust port **700** is provided at a corner portion of the second panel **30**. In the first embodiment, the exhaust port **700** is provided for the second panel **30**. However, this is only an example and should not be construed as limiting. The exhaust port **700** may be provided for the first panel **20**, or the second panel **30**, or a part of the hot glue forming the frame member **410**.

(39) The gas adsorbent **60** is arranged in the first space **510**. Specifically, the gas adsorbent **60** has an elongate shape and is arranged at a second end along the length of the second panel **30** (i.e., the left end in FIG. **4**) to extend along the width of the second panel **30**. That is to say, the gas adsorbent **60** is arranged at an end of the first space **510** (evacuated space **50**), thus making the gas adsorbent **60** less conspicuous. In addition, the gas adsorbent **60** is provided at some distance from the partition **420** and the air passage **600**. This reduces, when a gas is being exhausted from the first space **510**, the chances of the gas adsorbent **60** interfering with the gas exhaustion.
(40) The gas adsorbent **60** is used to adsorb an unnecessary gas (such as a residual gas). The unnecessary gas is emitted from the frame member **410** and the partition **420** when the frame member **410** and the partition **420** are heated, for example.
(41) The gas adsorbent **60** includes a getter. The getter is a material having the property of adsorbing molecules smaller in size than a predetermined one. The getter may be an evaporative getter, for example. The evaporative getter has the property of releasing adsorbed molecules when heated to a predetermined temperature (activation temperature) or more. This allows, even if the adsorption ability of the evaporative getter deteriorates, the evaporative getter to recover its adsorption ability by being heated to the activation temperature or more. The evaporative getter may be a zeolite or an ion-exchanged zeolite (such as a copper-ion-exchanged zeolite).
(42) The gas adsorbent **60** includes a powder of this getter. Specifically, the gas adsorbent **60** may be formed by applying a solution in which a powder of the getter is dispersed. This reduces the size of the gas adsorbent **60**, thus allowing the gas adsorbent **60** to be arranged even when the evacuated space **50** is narrow.
(43) The plurality of pillars **70** is used to maintain a predetermined gap between the first panel **20** and the second panel **30**. That is to say, the plurality of pillars **70** serves spacers to maintain the gap distance between the first panel **20** and the second panel **30** at a desired value.
(44) The plurality of pillars **70** are placed in the first space **510**. Specifically, the plurality of pillars **70** are arranged at the intersections of a rectangular (either square or rectangular) grid. The interval between the plurality of pillars **70** may be at least 2 cm, for example. Note that the dimensions, number, interval, and arrangement pattern of the pillars **70** may be selected appropriately.
(45) The pillars **70** may be made of a transparent material. However, this is only an example and should not be construed as limiting. Alternatively, the pillars **70** may also be made of an opaque material if the dimensions of the pillars **70** are sufficiently small. The material for the pillars **70** is selected to prevent the pillars **70** from being deformed in any of the bonding step, the evacuation step, or the sealing step to be described later. For example, the material for the pillars **70** is selected to have a softening point (softening temperature) higher than the first softening point of the first hot glue and the second softening point of the second hot glue.
(46) The closing member **80** is used to reduce the chances of dust, dirt or any other foreign particles entering the second space **520** through the exhaust port **700**. In the first embodiment, the closing member **80** may be a lid **81** provided on the surface of the exhaust port **700** of the first panel **20** or the second panel **30**.
(47) Providing such a closing member **80** for the exhaust port **700** reduces the chances of dust, dirt or any other foreign particles entering the second space **520** through the exhaust port **700**. This reduces the chances of the dust, dirt or any other foreign particles that have entered either the exhaust port **700** or the second space **520** adversely affecting the appearance of the glass panel unit **10**. Optionally, such a closing member **80** may be omitted.
(48) Next, a method for manufacturing the glass panel unit **10** according to the first embodiment will be described with reference to FIGS. **5-11**. The glass panel unit manufacturing method according to the first embodiment includes a glue arrangement step, a pillar placement step, an assembly forming step, a bonding step, an evacuation step, and a sealing step. Optionally, the manufacturing method may further include other additional process steps as needed. These manufacturing process steps will be described one by one.

(49) According to the first embodiment, first, a substrate forming step is performed, although not shown in any of the drawings. The substrate forming step is the process step of forming the first panel **20** and the second panel **30**. Specifically, the substrate forming step may include, for example, making the first panel **20** and the second panel **30**. Optionally, the substrate forming step may include cleaning the first panel **20** and the second panel **30** as needed.
(50) Next, the step of providing the exhaust port **700** is performed, in this process step, the exhaust port **700** is provided through the second panel **30**. Alternatively, the exhaust port **700** may be provided through the first panel **20** or may be provided through a hot glue to be arranged in the glue arrangement step to be described later. That is to say, the exhaust port **700** is provided through at least one of the first panel **20**, the second panel **30**, or the hot glue.
(51) Next, as shown in FIG. **5**, the glue arrangement step is performed. The glue arrangement step is the step of arranging a hot glue on either the first panel **20** or the second panel **30**. Specifically, the glue arrangement step includes forming a frame member **410** and a partition **420** on the second panel **30**. The glue arrangement step includes applying, with a dispenser, for example, a material for the frame member **410** (i.e., a first hot glue) and a material for the partition **420** (i.e., a second hot glue) onto the second panel **30** (i.e., the first surface of the second glass pane **31**). In the first embodiment, in the glue arrangement step, the air passage **600** is formed.
(52) Optionally, the glue arrangement step may include drying and pre-baking the material for the frame member **410** and the material for the partition **420**. For example, the second panel **30** on which the material for the frame member **410** and the material for the partition **420** are applied may be heated. If necessary, the first panel **20** may also be heated alone with the second panel **30**. That is to say, the first panel **20** and the second panel **30** may be heated under the same condition. Such prebaking may be omitted.
(53) Next, the pillar placement step is performed. The pillar placement step is the step of placing pillars **70** on either the first panel **20** or the second panel **30**. Specifically, the pillar placement step includes forming a plurality of pillars **70** in advance and placing, using a chip mounter, for example, the plurality of pillars **70** at predetermined positions on the second panel **30**. The plurality of pillars **70** may be formed by photolithographic and etching techniques. In that case, the plurality of pillars **70** may be made of a photocurable material, for example. Alternatively, the plurality of pillars **70** may also be formed by a known thin film forming technique. When measured from the upper surface of the second panel **30**, the height of the pillars **70** mounted on the second panel **30** is lower than the height of the hot glue mounted on the second panel **30**.
(54) Next, the gas adsorbent forming step is performed. Specifically, the gas adsorbent forming step includes forming a gas adsorbent **60** by applying a solution in which a powder of a getter is dispersed onto a predetermined region on the second panel **30** and drying the solution. Note that the glue arrangement step, the pillar placement step, and the gas adsorbent forming step may be performed in any arbitrary order.
(55) Next, the assembly forming step is performed. As shown in FIG. **6**, the assembly forming step includes forming an assembly **100** (see FIGS. **3** and **4**) by arranging the second panel **30** such that the second panel **30** faces the first panel **20**.
(56) The first panel **20** and the second panel **30** are arranged such that the first surface of the first glass pane **21** and the first surface of the second glass pane **31** are parallel to each other and face each other and laid one on top of the other. Performing this assembly forming step brings the hot glue into contact with the first panel **20** and the second panel **30**, thus forming the assembly **100** shown in FIGS. **3** and **4**.
(57) Next, the assembly mounting step is performed. The assembly mounting step includes mounting the assembly **100** on a supporting stage (not shown) of a melting furnace (not shown) as shown in FIG. **7**. In this case, only one supporting stage may be arranged in the melting furnace or a plurality of supporting stages may be arranged as a set in the melting furnace. In the latter case, the respective upper surfaces of the supporting stages are level with each other. The upper surface

of the supporting stage(s) is suitably a horizontal plane.

(58) In addition, a vacuum pump is connected to the assembly **100** via an exhaust pipe **810** and a sealing head **820**. The exhaust pipe **810** is bonded to the second panel **30** such that the inside of the exhaust pipe **810** communicates with the exhaust port **700**, for example. Then, the sealing head **820** is attached to the exhaust pipe **810**. In this manner, a suction port of the vacuum pump is connected to the exhaust port **700**.

(59) Next, the bonding step (first melting step) is performed. The bonding step includes heating the assembly **100** and melting the hot glue to bond the first panel **20** and the second panel **30** together with the hot glue and thereby form an internal space **500**. The internal space **500** is a space surrounded, except the exhaust port **700**, with the first panel **20**, the second panel **30**, and the hot glue melted.

(60) The bonding step and the evacuation step and sealing step following the bonding step are performed with the assembly **100** still loaded in the melting furnace.

(61) The bonding step includes hermetically bonding the first panel **20** and the second panel **30** together by once melting the first hot glue at a predetermined temperature (first melting temperature) Tm**1** equal to or higher than a first softening point. The bonding step is divided into a first temperature raising step, a first temperature maintaining step, and a first temperature lowering step according to temperature variations.

(62) The first temperature raising step is the process step performed for the time period t**1** shown in FIG. **8** and including raising the temperature in the melting furnace from ordinary temperature to the first melting temperature Tm**1**.

(63) The first temperature maintaining step is the process step performed for the time period t**2** shown in FIG. **8** and including maintaining the temperature in the melting furnace at the first melting temperature Tm**1**.

(64) The first temperature lowering step is the process step performed for the time period t**3** shown in FIG. **8** and including lowering the temperature in the melting furnace from the first melting temperature Tm**1** to a predetermined temperature (exhaustion temperature) Te to be described later.

(65) The first panel **20** and the second panel are loaded into the melting furnace and are heated at the first melting temperature Tm**1** for a predetermined amount of time (first melting time) t**2** in the first temperature maintaining step as shown in FIG. **8**.

(66) The first melting temperature Tm**1** and the first melting time t**2** are defined such that the first panel **20** and the second panel **30** are hermetically bonded together with the hot glue of the frame member **410** but that the air passage **600** is not closed by the partition **420**. That is to say, the lower limit of the first melting temperature Tm**1** is the first softening point, but the upper limit of the first melting temperature Tm**1** is set to prevent the air passage **600** from being closed by the partition **420**. For example, if the first softening point and the second softening point are 434° C., the first melting temperature Tm**1** is set at 440° C. Also, the first melting time t**2** may be 10 minutes, for example. Note that in the bonding step, a gas is emitted from the frame member **410** but is adsorbed into the gas adsorbent **60**.

(67) In the first temperature lowering step to be performed after the first temperature maintaining step, the first hot glue that has been once melted is going to cure, so are the first panel **20** and the second panel **30**.

(68) In this state, the pressure in the internal space **500** has not been reduced yet, and therefore, is as high as the pressure in the external environment. Thus, the first panel **20** and the second panel **30** do not receive force that causes the first and second panels **20**, **30** to approach each other under the atmospheric pressure. Therefore, the first panel **20** and the second panel **30** are not close to each other and the pillars **70** mounted on the second panel **30** are out of contact with the first panel **20**. This makes the pillars **70** easily displaceable on the second panel **30**.

(69) Next, the evacuation step is performed. The evacuation step includes reducing pressure in the internal space **500** by evacuation that involves exhausting, with predetermined suction power, a gas

from the internal space **500** via parallel channels **960** and the exhaust port **700** as shown in FIG. **9**A.
(70) The exhaustion may be carried out using vacuum pumps (**910**, **930** as will be described later). The vacuum pumps exhaust the gas from the internal space **500** via a manifold **900**, the sealing heads **820**, the exhaust pipes **810**, and the exhaust port **700**. The plurality of sealing heads **820** are connected to the manifold **900** via vent pipes.
(71) In the first embodiment, a low-vacuum pump **910** such as a rotary pump and a high-vacuum pump **930** such as a turbo molecular pump may be used as the vacuum pumps. Optionally, the manifold **900** or the vent pipes may be provided with a Penning gauge (for high vacuum) and a Pirani gauge (for low vacuum) as appropriate to measure the pressure.
(72) The parallel channels **960** include a plurality of channels that are connected together in parallel (namely, a first channel **961** and a second channel **962** to be described later) and one or two or more on-off valves (namely, a first valve **921** and a second valve **922** to be described later) provided for one or two or more channels out of the plurality of channels. The total channel area of the parallel channels **960** (i.e., the total of the respective cross-sectional areas of the one or two or more channels, of which the on-off valves are opened) may be changed according to a combination of the opened and closed states of the on-off valves.
(73) Specifically, the parallel channels **960** include: a first channel **961** for selectively allowing the gas to flow, or shutting off the gas, through a channel having a first channel area; and a second channel **962** for selectively allowing the gas to flow, or shutting off the gas, through a channel having a second channel area larger than the first channel area. The first channel **961** and the second channel **962** are provided in parallel.
(74) The low-vacuum pump **910** is connected to the manifold **900** via the parallel channels **960**. The first channel **961** has a first valve **921** provided on the way and the second channel **962** has a second valve **922** on the way. The first valve **921** and the second valve **922** together form a low-vacuum valve set **920**.
(75) The high-vacuum pump **930** is connected to the manifold **900** via a vent pipe having a high-vacuum valve **940** on the way. Optionally, although not shown, yet another relief valve may be connected to the manifold **900** via another channel.
(76) Also, the low-vacuum pump **910** and the high-vacuum pump **930** are connected together via a vent pipe having an auxiliary valve **950** on the way. Optionally, the low-vacuum pump **910** and the high-vacuum pump **930** do not have to be connected together via the vent pipe.
(77) The first valve **921** may be operated to selectively allow the gas to flow, or shut off the gas, through the first channel **961**. The first valve **921** is an on-off valve. The channel area of the first valve **921** opened is the first channel area. The channel area of the rest of the first channel **961** other than the first valve **921** is equal to or greater than the first channel area. The second valve **922** may be operated to selectively allow the gas to flow, or shut off the gas, through the second channel **962**. The second valve **922** is an on-off valve. The channel area of the second valve **922** opened is the second channel area. The channel area of the rest of the second channel **962** other than the second valve **922** is equal to or greater than the second channel area. Each of the high-vacuum valve **940** and the auxiliary valve **950** may also be operated to selectively allow the gas to flow, or shut off the gas, through a channel having a predetermined channel area.
(78) The low-vacuum pump **910** has predetermined suction power. Meanwhile, the high-vacuum pump **930** may have different predetermined suction power from the low-vacuum pump **910**. However, this is only an example of the present disclosure and should not be construed as limiting. Alternatively, the high-vacuum pump **930** may have the same suction power as the low-vacuum pump **910**.
(79) As shown in FIG. **10**, the evacuation step may be roughly divided into a low vacuum forming step to be performed by the low-vacuum pump **910** (which is a step designated by the reference signs t**11** and t**12** and which will be hereinafter simply designated by t**11**, t**12**) and a high vacuum

forming step to be performed by the high-vacuum pump **930** (which is a step designated by the reference sign t**20** and which will be hereinafter simply designated by t**20**). The low vacuum forming step is performed first, which is followed by the high vacuum forming step.
(80) The low vacuum forming step is further subdivided into a first evacuation step t**11** to be performed first and a second evacuation step t**12** to be performed next to the first evacuation step t**11**.
(81) In the first evacuation step t**11**, the gas is allowed to flow with the total channel area of the parallel channels **960** set at a first total channel area. Specifically, in the first evacuation step t**11**, the first valve **921** is opened to allow the gas to flow through the first channel **961** and the second valve **922** is closed to shut off the gas through the second channel **962** (see FIG. **9**A).
(82) In the second evacuation step t**12**, the gas is allowed to flow with the total channel area of the parallel channels **960** set at a second total channel area which is larger than the first total channel area. Specifically, in the second evacuation step t**12**, the first valve **921** is closed to shut off the gas through the first channel **961** and the second valve **922** is opened to allow the gas to flow through the second channel **962** (see FIG. **9**A). Note that the high-vacuum valve **940** is closed.
(83) Since the first channel area of the first channel **961** is smaller than the second channel area of the second channel **962**, the suction power applied from the exhaust port **700** is less in the first evacuation step t**11** to be performed first than in the second evacuation step t**12**. Thus, when the gas is exhausted from the internal space **500** through the exhaust port **700**, the air flowing from the vicinity of the exhaust port **700** of the internal space **500** toward the exhaust port **700** may have a lower flow velocity. Specifically, the air flowing from the vicinity of the exhaust port **700** of the internal space **500** toward the exhaust port **700** may have a lower flow velocity than in a situation where the gas is exhausted with the first valve **921** closed and the second valve **922** opened in the first evacuation step t**11**. This reduces the chances of the pillars **70** which are not firmly clamped between the first panel **20** and the second panel **30** being displaced by the air flowing toward the exhaust port **700**.
(84) When the gas has been exhausted to a certain degree from the internal space **500** in the first evacuation step t**11**, the first evacuation step t**11** will be ended and the second variation step t**12** will be started instead.
(85) In the first embodiment, the first evacuation step t**11** may be performed for one minute and the second evacuation step t**12** may also be performed for one minute. However, this duration is only an example and should not be construed as limiting.
(86) This allows a predetermined degree of vacuum to be achieved in a short time with the displacement of the pillars **70** by the air flowing toward the exhaust port **700** reduced.
(87) For your reference, FIG. **11** shows how the pressure varies with time in a situation where the low vacuum forming step (which is a step designated by t**10** and will be hereinafter simply designated by t**10**) is performed with the second valve **922** opened and the first valve **921** closed. In that case, the duration of the low vacuum forming step t**10** may be as short as less than two minutes but the pillars **70** tend to be displaced by the air flowing toward the exhaust port **700**.
(88) After the low vacuum forming step has been performed, a high vacuum forming step t**20** is performed with the first valve **921** and the second valve **922** closed and with the high-vacuum valve **940** opened.
(89) The evacuation step is subdivided into a first temperature lowering step, a second temperature maintaining step, a second temperature raising step, a third temperature maintaining step, and a second temperature lowering step according to temperature variations.
(90) The first temperature lowering step of the evacuation step partially overlaps with the first temperature lowering step of the bonding step preceding the evacuation step. That is to say, in the middle of the first temperature lowering step of the bonding step, the evacuation step is started. In particular, the evacuation step is suitably started when the temperature of the assembly **100** is equal to or higher than the softening point of the hot glue. Alternatively, the evacuation step may begin

during the next second temperature maintaining step, not during the first temperature lowering step.
(91) The second temperature maintaining step is the process step performed for the time period t**4** shown in FIG. **8** and including maintaining the temperature in the melting furnace at an exhaustion temperature Te. The second temperature maintaining step includes exhausting a gas from the first space **510** via the air passage **600**, the second space **520**, and the exhaust port **700**.
(92) The exhaustion temperature Te is set at a temperature higher than the getter activation temperature (of 350° C., for example) of the gas adsorbent **60** but lower than the first softening point and the second softening point (of 434° C.). For example, the exhaustion temperature Te may be 390° C.
(93) This prevents the frame member **410** and the partition **420** from being deformed. In addition, this activates the getter of the gas adsorbent **60**, thus causing the molecules (gas) adsorbed into the getter to be released from the getter. Then, the molecules (i.e., the gas) released from the getter are exhausted via the first space **510**, the air passage **600**, the second space **520**, and the exhaust port **700**. Thus, the evacuation step allows the gas adsorbent **60** to recover its adsorption ability.
(94) The exhaustion time t**4** is set to create an evacuated space **50** with a desired degree of vacuum (e.g., a degree of vacuum of 0.1 Pa or less). The exhaustion time t**4** may be set at 120 minutes, for example. Note that the degree of vacuum of the evacuated space **50** is not particularly limited.
(95) The second temperature raising step, the third temperature maintaining step, and the second temperature lowering step following the second temperature maintaining step overlap with the sealing step following the second temperature maintaining step. That is to say, in the second temperature raising step, the third temperature maintaining step, and the second temperature lowering step, the evacuation step and the sealing step are performed in parallel.
(96) Next, the sealing step (second melting step) is performed. The sealing step includes creating a hermetically sealed evacuated space **50** by closing the exhaust port **700** and sealing the internal space **500** while maintaining the reduced pressure in the internal space **500**.
(97) The second temperature raising step is the process step performed for the time period t**5** shown in FIG. **8** and including raising the temperature in the melting furnace from the exhaustion temperature Te to a predetermined temperature (second melting temperature) Tm**2**.
(98) The third temperature maintaining step is the process step performed for the time period t**6** shown in FIG. **8** and including maintaining the temperature in the melting furnace at the second melting temperature Tm**2**.
(99) The third temperature maintaining step includes melting the second hot glue once at a predetermined temperature Tm**2** equal to or higher than the second softening point to deform the partition **420** and thereby form a boundary wall **42** closing the air passage **600**. Specifically, the first panel **20** and the second panel **30** are heated at the second melting temperature Tm**2** for a predetermined amount of time (second melting time) t**6** in the melting furnace.
(100) The partition **420** contains the second hot glue. Thus, melting the second hot glue once at the predetermined temperature (second melting temperature) Tm**2** that is equal to or higher than the second softening point allows the partition **420** to be deformed into the boundary wall **42** shown in FIGS. **1** and **2**. Note that the first melting temperature Tm**1** is set at a temperature lower than the second melting temperature Tm**2**. This prevents, when the first panel **20** and the second panel **30** are bonded together with the frame member **410**, the partition **420** from being deformed to close the air passage **600**.
(101) The second melting temperature Tm**2** and the second melting time t**6** are set such that the second hot glue softens to form the boundary wall **42** that closes the air passage **600**. The lower limit of the second melting temperature Tm**2** is the second softening point (of 434° C.). Nevertheless, the sealing step is performed for the purpose of deforming the partition **420** unlike the bonding step. Thus, the second melting temperature Tm**2** is set at a temperature higher than the first melting temperature (of 440° C.) Tm**1**. The second melting temperature Tm**2** may be set at 460° C., for example. Also, the second melting time t**6** may be 30 minutes, for example.

(102) The second temperature lowering step is the process step performed for the time period t**7** shown in FIG. **8** and includes lowering the temperature in the melting furnace from the second melting temperature Tm**2** to ordinary temperature.
(103) After the second temperature lowering step has been performed, the glass panel unit **10** will be unloaded from the melting furnace.
(104) The glass panel unit **10** thus obtained includes the first panel **20**, the second panel **30**, the seal **40**, the evacuated space **50**, the second space **520**, the gas adsorbent **60**, the plurality of pillars **70**, and the closing member **80** as shown in FIG. **2**.
(105) The evacuated space **50** is formed by exhausting a gas from the first space **510** via the second space **520** and the exhaust port **700** as described above. In other words, the evacuated space **50** is the first space **510**, of which the degree of pressure reduction is equal to or less than a predetermined value. The predetermined value may be 0.1 Pa, for example. The evacuated space **50** is hermetically closed completely by the first panel **20**, the second panel **30**, and the seal **40**, and therefore, is separated from the second space **520** and the exhaust port **700**.
(106) The seal **40** not only surrounds the evacuated space **50** entirely but also hermetically bonds the first panel **20** and the second panel **30** together. The seal **40** has the shape of a frame including a first part **41** and a second part (boundary wall **42**). The first part **41** is a portion, corresponding to the evacuated space **50**, of the frame member **410**. That is to say, the first part **41** is a portion, facing the evacuated space **50**, of the frame member **410**. The second part is the boundary wall **42** obtained by deforming the partition **420**.
(107) FIG. **9**B illustrates another exemplary exhaust system according to the first embodiment. In this example, the first channel **961** includes a throttle valve **963** having the first channel area and a first valve **923** for selectively allowing the gas to flow, or shutting off the gas, through the first channel **961**. The channel area of the rest of the first channel **961** (including the first valve **923**) other than the throttle valve **963** is larger than the first channel area. The second channel **962** includes a throttle valve **964** having the second channel area and a second valve **924** for selectively allowing the gas to flow, or shutting off the gas, through the second channel **962**. The channel area of the rest of the second channel **962** (including the second valve **924**) other than the throttle valve **964** is larger than the second channel area. This alternative exhaust system shown in FIG. **9**B may also achieve the same advantages as the exhaust system shown in FIG. **9**A.
(108) Next, a method for manufacturing a glass panel unit **10** according to a second embodiment will be described with reference to FIG. **12**. The method for manufacturing the glass panel unit **10** according to the second embodiment is mostly the same as the method for manufacturing the glass panel unit **10** according to the first embodiment. Thus, in the following description, their common features will not be described all over again to avoid redundancies.
(109) In the first embodiment described above, the gas in the internal space **500** is exhausted through the parallel channels **960** in the evacuation step as shown in FIGS. **9**A and **9**B. In the second embodiment, on the other hand, the gas in the internal space **500** is exhausted in the evacuation step by a pump (low-vacuum pump **910**) for suctioning the gas with predetermined suction power via predetermined channels **970** as shown in FIG. **12**.
(110) The predetermined channels **970** include: an exhaust channel **971** which is located on a path connecting the low-vacuum pump **910** to the exhaust port **700** and which has an exhaust on-off valve **925**; and an air-introducing channel **972** which is located on a path connecting the low-vacuum pump **910** to the air and which has an air-introducing on-off valve **926**.
(111) The evacuation step includes: a first evacuation step to be performed first; and a second evacuation step to be performed next to the first evacuation step.
(112) In the first evacuation step, the exhaust on-off valve **925** is opened and the air-introducing on-off valve **926** is also opened. In this case, settings are made such that even when the exhaust on-off valve **925** is opened and the air-introducing on-off valve **926** is also opened, the air introduced through the air-introducing channel **972** does not flow toward the exhaust port **700** but flows

toward the low-vacuum pump **910**.
(113) In the second evacuation step, the exhaust on-off valve **925** is opened and the air-introducing on-off valve **926** is closed.
(114) In the second embodiment, the pressure in the internal space also varies with time in the same way as in FIG. **10** showing how the pressure in the internal space varies with time in the first embodiment.
(115) As shown in FIG. **10**, in the first evacuation step t**11**, the suction power applied from the exhaust port **700** is less than in the second evacuation step t**12**. That is to say, in the first evacuation step t**11**, the air-introducing on-off valve **926** is also opened so that the air is introduced through the air-introducing channel **972** and suctioned into the low-vacuum pump **910**. This decreases, compared to a situation where the air-introducing on-off valve **926** is closed, the suction power applied by the low-vacuum pump **910** for suctioning the gas from the exhaust port **700** and also decreases the flow velocity of the air flowing from the vicinity of the exhaust port **700** of the internal space **500** toward the exhaust port **700**. This reduces the chances of the pillars **70** which are not firmly clamped between the first panel **20** and the second panel **30** being displaced by the air flowing toward the exhaust port **700**.
(116) Next, a method for manufacturing a glass panel unit **10** according to a third embodiment will be described with reference to FIGS. **13** and **14**. The method for manufacturing the glass panel unit **10** according to the third embodiment is mostly the same as the method for manufacturing the glass panel unit **10** according to the first embodiment. Thus, in the following description, their common features will not be described all over again to avoid redundancies.
(117) In the first embodiment described above, the low-vacuum valves **920** include the first valve **921** and the second valve **922** as shown in FIG. **9**A. In the third embodiment, on the other hand, only one low-vacuum valve **920** is provided. The low-vacuum valve **920** according to the third embodiment, as well as the second valve **922** according to the first embodiment, has the second channel area.
(118) As shown in FIG. **14**, in the first evacuation step t**11** and the second evacuation step t**12**, the low-vacuum valve **920** has its opened and closed states controlled (subjected to duty control).
(119) A first ratio of a valve open duration of the low-vacuum valve **920** during the first evacuation step t**11** to an overall running time of the first evacuation step t**11** is smaller than a second ratio of a valve open duration of the low-vacuum valve **920** during the second evacuation step t**12** to an overall running time of the second evacuation step t**12**.
(120) Specifically, in the first evacuation step t**11**, the low-vacuum valve **920** may be opened for 0.1 seconds, closed for 0.9 seconds next, and then this cycle is repeated at regular time intervals of 5 seconds, 10 seconds, or 1 minute, for example. In that case, the first ratio becomes 0.1.
(121) Throughout the second evacuation step t**12**, the low-vacuum valve **920** is opened. In that case, the second ratio becomes 1.0.
(122) Note that the valve open duration and valve closed duration of the low-vacuum valve **920** and the first and second ratios are not limited to any particular values.
(123) According to the third embodiment, in the first evacuation step t**11** to be performed first, the suction power applied from the exhaust port **700** is less than in the second evacuation step t**12**. This decreases the flow velocity of the air flowing from the vicinity of the exhaust port **700** of the internal space **500** toward the exhaust port **700** while the gas is being exhausted from the internal space **500** through the exhaust port **700**. This reduces the chances of the pillars **70** which are not firmly clamped between the first panel **20** and the second panel **30** being displaced by the air flowing toward the exhaust port **700**.
(124) This allows a predetermined degree of vacuum to be achieved in a short time with the displacement of the pillars **70** by the air flowing toward the exhaust port **700** reduced.
(125) Next, a method for manufacturing a glass panel unit **10** according to a fourth embodiment will be described with reference to FIG. **15**. The method for manufacturing the glass panel unit **10**

according to the fourth embodiment is mostly the same as the method for manufacturing the glass panel unit **10** according to the first embodiment. Thus, in the following description, their common features will not be described all over again to avoid redundancies.
(126) In the fourth embodiment, the gas is exhausted in the same way as shown in FIG. **11** with the second valve **922** opened throughout the low vacuum forming step t**10**.
(127) At this time, the exhaust path is specially designed to reduce the chances of the pillars **70** which are not firmly clamped between the first panel **20** and the second panel **30** being displaced by the air flowing toward the exhaust port **700**.
(128) Specifically, as shown in FIG. **15**A, the partition **420** is formed along almost the entire width of the assembly **100** and air passages **600** are provided at multiple positions along the length of the partition **420**. This increases the total volume of the air passing through the respective air passage **600**, thus decreasing the flow velocity of the air colliding against the pillars **70** in the vicinity of the air passage **600** while passing through the air passage **600**. This reduces the chances of the pillars **70** which are not firmly clamped between the first panel **20** and the second panel **30** being displaced by the air flowing toward the exhaust port **700**.
(129) In addition, as shown in FIG. **15**B, the exhaust path running from the air passage **600** between the partition **420** and the frame member **410** through the exhaust port **700** is extended. In this fourth embodiment, the exhaust path running from the air passage **600** through the exhaust port **700** is extended by providing an additional partition **420**. Thus, the extended exhaust path increases the drag against the air flowing through the exhaust path and decreases the flow velocity of the air passing through the respective air passages **600**. This reduces the chances of the pillars **70** which are not firmly clamped between the first panel **20** and the second panel **30** being displaced by the air flowing toward the exhaust port **700**.
(130) Next, variations of the first to fourth embodiments will be described.
(131) In the embodiments described above, the glass panel unit **10** has a rectangular shape. However, this is only an example and should not be construed as limiting. Alternatively, the glass panel unit **10** may also have a circular, polygonal, or any other desired shape. That is to say, the first panel **20**, the second panel **30**, and the seal **40** do not have to be rectangular but may also have a circular, polygonal, or any other desired shape. In addition, the respective shapes of the first panel **20**, the second panel **30**, the frame member **410**, and the boundary wall **42** do not have to be the ones used in the embodiment described above, but may also be any other shapes that allow a glass panel unit **10** of a desired shape to be obtained. Note that the shape and dimensions of the glass panel unit **10** may be determined according to the intended use of the glass panel unit **10**.
(132) Neither the first surface nor second surface of the first glass pane **21** of the first panel **20** has to be a plane. Likewise, neither the first surface nor second surface of the second glass pane **31** of the second panel **30** has to be a plane.
(133) The first glass pane **21** of the first panel **20** and the second glass pane **31** of the second panel **30** do not have to have the same planar shape and planar dimensions. The first glass pane **21** and the second glass pane **31** do not have to have the same thickness, either. In addition, the first glass pane **21** and the second glass pane **31** do not have to be made of the same material, either.
(134) The seal **40** does not have to have the same planar shape as the first panel **20** and the second panel **30**. Likewise, the frame member **410** does not have to have the same planar shape as the first panel **20** and the second panel **30**, either.
(135) Optionally, the first panel **20** may include a coating having desired physical properties and formed on the second surface of the first glass pane **21**. Alternatively, the first panel **20** may have no coatings **22**. That is to say, the first panel **20** may consist of the first glass pane **21** alone.
(136) Furthermore, the second panel **30** may include a coating with desired physical properties. The coating may include at least one of thin films respectively formed on the first and second surfaces of the second glass pane **31**, for example. The coating may be an infrared reflective film or ultraviolet reflective film that reflects light with a particular wavelength, for example.

(137) Furthermore, in the embodiments described above, the frame member **410** is made of the first hot glue. However, this is only an example and should not be construed as limiting. Alternatively, the frame member **410** may include not only the first hot glue but also a core material or any other material as well. That is to say, the frame member **410** needs to include at least the first hot glue. Furthermore, in the embodiments described above, the frame member **410** is formed to cover the second panel **30** almost entirely. However, this is only an example and should not be construed as limiting. Rather the frame member **410** needs to be formed to cover a predetermined area on the second panel **30**. That is to say, the frame member **410** does not have to be formed to cover almost the entire area on the second panel **30**.
(138) Furthermore, in the embodiments described above, the partition **420** is made of the second hot glue. However, this is only an example and should not be construed as limiting. Alternatively, the partition **420** may include not only the second hot glue but also a core material or any other material as well. That is to say, the partition **420** needs to include at least the second hot glue.
(139) In the embodiments described above, the internal space **500** is partitioned into a single first space **510** and a single second space **520**. Optionally, the internal space **500** may also be partitioned into one or more first spaces **510** and one or more second spaces **520**.
(140) The first glue and the second hot glue do not have to be glass frit but may also be a low-melting metal or a hot melt adhesive, for example.
(141) In the embodiments described above, a melting furnace is used to heat the frame member **410**, the gas adsorbent **60**, and the partition **420**. However, heating may be conducted by any appropriate heating means. The heating means may be a laser beam or a heat transfer plate connected to a heat source, for example.
(142) In the embodiments described above, the exhaust port **700** is provided through the second panel **30**. However, this is only an example and should not be construed as limiting. Alternatively, the exhaust port **700** may be provided through the first glass pane **21** of the first panel **20** or through the frame member **410**.
(143) As can be seen from the foregoing description of embodiments, a method for manufacturing a glass panel unit (**10**) according to a first aspect includes a glue arrangement step, a pillar placement step, an assembly forming step, a bonding step, an evacuation step, and a sealing step.
(144) The glue arrangement step includes arranging a hot glue on either a first panel (**20**) including a first glass pane (**21**) or a second panel (**30**) including a second glass pane (**31**).
(145) The pillar placement step includes placing pillars (**70**) on either the first panel (**20**) or the second panel (**30**).
(146) The assembly forming step includes forming an assembly (**100**) including the first panel (**20**), the second panel (**30**), the hot glue, and the pillars (**70**) and having an exhaust port (**700**) provided through at least one of the first panel (**20**), the second panel (**30**), or the hot glue by arranging the second panel (**30**) such that the second panel (**30**) faces the first panel (**20**).
(147) The bonding step includes heating the assembly (**100**) to melt the hot glue, bonding the first panel (**20**) and the second panel (**30**) together with the hot glue thus melted, and thereby forming an internal space (**500**) surrounded, except the exhaust port (**700**), with the first panel (**20**), the second panel (**30**), and the hot glue melted.
(148) The evacuation step includes reducing pressure in the internal space (**500**) by evacuation that involves exhausting, with predetermined suction power, a gas from the internal space (**500**) via parallel channels (**960**) and the exhaust port (**700**).
(149) The sealing step includes creating a hermetically sealed evacuated space (**50**) by closing the exhaust port (**700**) and sealing the internal space (**500**) while maintaining a reduced pressure in the internal space (**500**).
(150) The parallel channels (**960**) include: a plurality of channels (a first channel (**961**) and a second channel (**962**)) that are connected together in parallel; and one or two or more on-off valves (a first valve (**921**) and a second valve (**922**)) provided for one or two or more channels out of the

plurality of channels. A total channel area of the parallel channels (**960**) is changeable according to a combination of opened and closed states of the on-off valves.
(151) The evacuation step includes: a first evacuation step to be performed first; and a second evacuation step to be performed next to the first evacuation step.
(152) The first evacuation step includes allowing the gas to flow while setting the total channel area of the parallel channels (**960**) at a first total channel area.
(153) The second evacuation step includes allowing the gas to flow while setting the total channel area of the parallel channels (**960**) at a second total channel area that is larger than the first total channel area.
(154) A method for manufacturing a glass panel unit (**10**) according to the first aspect reduces the chances of the pillars (**70**) being displaced by the air flowing toward the exhaust port (**700**).
(155) A method for manufacturing a glass panel unit (**10**) according to a second aspect includes a glue arrangement step, a pillar placement step, an assembly forming step, a bonding step, an evacuation step, and a sealing step.
(156) The glue arrangement step includes arranging a hot glue on either a first panel (**20**) including a first glass pane (**21**) or a second panel (**30**) including a second glass pane (**31**).
(157) The pillar placement step includes placing pillars (**70**) on either the first panel (**20**) or the second panel (**30**).
(158) The assembly forming step includes forming an assembly (**100**) including the first panel (**20**), the second panel (**30**), the hot glue, and the pillars (**70**) and having an exhaust port (**700**) provided through at least one of the first panel (**20**), the second panel (**30**), or the hot glue by arranging the second panel (**30**) such that the second panel (**30**) faces the first panel (**20**).
(159) The bonding step includes heating the assembly (**100**) to melt the hot glue, bonding the first panel (**20**) and the second panel (**30**) together with the hot glue thus melted, and thereby forming an internal space (**500**) surrounded, except the exhaust port (**700**), with the first panel (**20**), the second panel (**30**), and the hot glue melted.
(160) The evacuation step includes reducing pressure in the internal space (**500**) by evacuation that involves exhausting a gas, using a pump (low-vacuum pump (**910**)) configured to suction the gas with predetermined suction power, from the internal space (**500**) via a predetermined channel (**970**) and the exhaust port (**700**).
(161) The sealing step includes creating a hermetically sealed evacuated space (**50**) by closing the exhaust port (**700**) and sealing the internal space (**500**) while maintaining a reduced pressure in the internal space (**500**).
(162) The predetermined channel includes: an exhaust channel (**971**) located on a path connecting the pump (low-vacuum pump (**910**)) to the exhaust port (**700**) and haying an exhaust on-off valve (**925**); and an air-introducing channel (**972**) located on a path connecting the pump (low-vacuum pump (**910**)) to the air and having an air-introducing on-off valve (**926**).
(163) The evacuation step includes: a first evacuation step to be performed first; and a second evacuation step to be performed next to the first evacuation step.
(164) The first evacuation step includes opening the exhaust on-off valve (**925**) and opening the air-introducing on-off valve (**926**).
(165) The second evacuation step includes opening the exhaust on-off valve (**925**) and closing the air-introducing on-off valve (**926**).
(166) A method for manufacturing a glass panel unit (**10**) according to the second aspect reduces the chances of the pillars (**70**) being displaced by the air flowing toward the exhaust port (**700**).
(167) A method for manufacturing a glass panel unit (**10**) according to a third aspect includes a glue arrangement step, a pillar placement step, an assembly forming step, a bonding step, an evacuation step, and a sealing step.
(168) The glue arrangement step includes arranging a hot glue on either a first panel (**20**) including a first glass pane (**21**) or a second panel (**30**) including a second glass pane (**31**).

(169) The pillar placement step includes placing pillars (**70**) on either the first panel (**20**) or the second panel (**30**).
(170) The assembly forming step includes forming an assembly (**100**) including the first panel (**20**), the second panel (**30**), the hot glue, and the pillars (**70**) and having an exhaust port (**700**) provided through at least one of the first panel (**20**), the second panel (**30**), or the hot glue by arranging the second panel (**30**) such that the second panel (**30**) faces the first panel (**20**).
(171) The bonding step includes heating the assembly (**100**) to melt the hot glue, bonding the first panel (**20**) and the second panel (**30**) together with the hot glue thus melted, and thereby forming an internal space (**500**) surrounded, except the exhaust port (**700**), with the first panel (**20**), the second panel (**30**), and the hot glue melted.
(172) The evacuation step includes reducing pressure in the internal space (**500**) by evacuation that involves exhausting, with predetermined suction power, a gas from the internal space (**500**) via a valve (low-vacuum valve (**920**)) and the exhaust port (**700**).
(173) The sealing step includes creating a hermetically sealed evacuated space (**50**) by closing the exhaust port (**700**) and sealing the internal space (**500**) while maintaining a reduced pressure in the internal space (**500**).
(174) The evacuation step includes: a first evacuation step to be performed first; and a second evacuation step to be performed next to the first evacuation step. The first evacuation step and the second evacuation step each include controlling opened and closed states of the valve.
(175) A first ratio of a valve open duration during the first evacuation step to an overall running time of the first evacuation step is smaller than a second ratio of a valve open duration during the second evacuation step to an overall running time of the second evacuation step. The valve open duration is a period of time for which the valve is opened.
(176) A method for manufacturing a glass panel unit (**10**) according to the third aspect reduces the chances of the pillars (**70**) being displaced by the air flowing toward the exhaust port (**700**).
REFERENCE SIGNS LIST
(177) **10** Glass Panel Unit
(178) **20** First Panel
(179) **21** First Glass Pane
(180) **30** Second Panel
(181) **31** Second Glass Pane
(182) **50** Evacuated Space
(183) **70** Pillar
(184) **100** Assembly
(185) **500** Internal Space
(186) **700** Exhaust Port

Claims

1. A method for manufacturing a glass panel unit, the method comprising: a glue arrangement step including arranging a hot glue on either a first panel or a second panel, the first panel including a first glass pane, the second panel including a second glass pane; a pillar placement step including placing pillars on either the first panel or the second panel; an assembly forming step including forming an assembly including the first panel, the second panel, the hot glue, and the pillars and having an exhaust port provided through at least one of the first panel, the second panel, or the hot glue by arranging the second panel such that the second panel faces the first panel; a bonding step including heating the assembly to melt the hot glue, bonding the first panel and the second panel together with the hot glue thus melted, and thereby forming an internal space surrounded, except the exhaust port, with the first panel, the second panel, and the hot glue melted; an evacuation step including reducing pressure in the internal space by evacuation that involves exhausting by a low-

vacuum pump, with predetermined suction power, a gas from the internal space via parallel channels and the exhaust port and exhausting by a high-vacuum pump, with predetermined suction power, a gas from the internal space via a vent pipe having a high-vacuum valve on a way and the exhaust port; and a sealing step including creating a hermetically sealed evacuated space by closing the exhaust port and sealing the internal space while maintaining a reduced pressure in the internal space, the parallel channels including: a plurality of channels that are connected together in parallel; and one or two or more on-off valves provided for one or two or more channels out of the plurality of channels, a total channel area of the parallel channels being changeable according to a combination of opened and closed states of the on-off valves, the evacuation step including: a first evacuation step to be performed first by the low-vacuum pump; a second evacuation step to be performed by the low-vacuum pump next to the first evacuation step; and a high vacuum forming step to be performed by the high-vacuum pump next to the second evacuation step, the first evacuation step including allowing the gas to flow while setting the total channel area of the parallel channels at a first total channel area, the second evacuation step including allowing the gas to flow while setting the total channel area of the parallel channels at a second total channel area that is larger than the first total channel area, and the high vacuum forming step including allowing the gas to flow with the on-off valves closed and with the high-vacuum valve opened.

2. A method for manufacturing a glass panel unit, the method comprising: a glue arrangement step including arranging a hot glue on either a first panel or a second panel, the first panel including a first glass pane, the second panel including a second glass pane; a pillar placement step including placing pillars on either the first panel or the second panel; an assembly forming step including forming an assembly including the first panel, the second panel, the hot glue, and the pillars and having an exhaust port provided through at least one of the first panel, the second panel, or the hot glue by arranging the second panel such that the second panel faces the first panel; a bonding step including heating the assembly to melt the hot glue, bonding the first panel and the second panel together with the hot glue thus melted, and thereby forming an internal space surrounded, except the exhaust port, with the first panel, the second panel, and the hot glue melted; an evacuation step including reducing pressure in the internal space by evacuation that involves exhausting a gas, using a pump configured to suction the gas with predetermined suction power, from the internal space via a predetermined channel and the exhaust port; and a sealing step including creating a hermetically sealed evacuated space by closing the exhaust port and sealing the internal space while maintaining a reduced pressure in the internal space, the predetermined channel including: an exhaust channel located on a path connecting the pump to the exhaust port and having an exhaust on-off valve; and an air-introducing channel located on a path connecting the pump to the air and having an air-introducing on-off valve, the evacuation step including: a first evacuation step to be performed first; and a second evacuation step to be performed next to the first evacuation step, the first evacuation step including opening the exhaust on-off valve and opening the air-introducing on-off valve, the second evacuation step including maintaining the exhaust on-off valve open and closing the air-introducing on-off valve.

3. A method for manufacturing a glass panel unit, the method comprising: a glue arrangement step including arranging a hot glue on either a first panel or a second panel, the first panel including a first glass pane, the second panel including a second glass pane; a pillar placement step including placing pillars on either the first panel or the second panel; an assembly forming step including forming an assembly including the first panel, the second panel, the hot glue, and the pillars and having an exhaust port provided through at least one of the first panel, the second panel, or the hot glue by arranging the second panel such that the second panel faces the first panel; a bonding step including heating the assembly to melt the hot glue, bonding the first panel and the second panel together with the hot glue thus melted, and thereby forming an internal space surrounded, except the exhaust port, with the first panel, the second panel, and the hot glue melted; an evacuation step including reducing pressure in the internal space by evacuation that involves exhausting by a low-

vacuum pump, with predetermined suction power, a gas from the internal space via a low-vacuum valve and the exhaust port and exhausting by a high-vacuum pump, with predetermined suction power, a gas from the internal space via a high-vacuum valve and the exhaust port; and a sealing step including creating a hermetically sealed evacuated space by closing the exhaust port and sealing the internal space while maintaining a reduced pressure in the internal space, the evacuation step including: a first evacuation step to be performed first; and a second evacuation step to be performed next to the first evacuation step, the first evacuation step and the second evacuation step each including controlling opened and closed states of the low-vacuum valve, a first ratio of a low-vacuum valve open duration during the first evacuation step to an overall running time of the first evacuation step being smaller than a second ratio of a low-vacuum valve open duration during the second evacuation step to an overall running time of the second evacuation step, the low-vacuum valve open duration being a period of time for which the low-vacuum valve is opened.

4. A method for manufacturing a glass panel unit, the method comprising: a glue arrangement step including arranging a hot glue on either a first panel or a second panel, the first panel including a first glass pane, the second panel including a second glass pane; a pillar placement step including placing pillars on either the first panel or the second panel; an assembly forming step including forming an assembly including the first panel, the second panel, the hot glue, and the pillars and having an exhaust port provided through at least one of the first panel, the second panel, or the hot glue by arranging the second panel such that the second panel faces the first panel; a bonding step including heating the assembly to melt the hot glue, bonding the first panel and the second panel together with the hot glue thus melted, and thereby forming an internal space surrounded, except the exhaust port, with the first panel, the second panel, and the hot glue melted; an evacuation step including reducing pressure in the internal space by evacuation that involves: exhausting by a low-vacuum pump, with predetermined suction power, a gas from the internal space via a low-vacuum valve and the exhaust port; and exhausting by a high-vacuum pump, with predetermined suction power, a gas from the internal space via a high-vacuum valve and the exhaust port; and a sealing step including creating a hermetically sealed evacuated space by closing the exhaust port and sealing the internal space while maintaining a reduced pressure in the internal space, the evacuation step including: a first evacuation step to be performed first; and a second evacuation step to be performed after the first evacuation step, the first evacuation step including controlling the low-vacuum valve such that the low-vacuum valve is opened and closed multiple times, the second evacuation step including keeping the low-vacuum valve in the opened state.